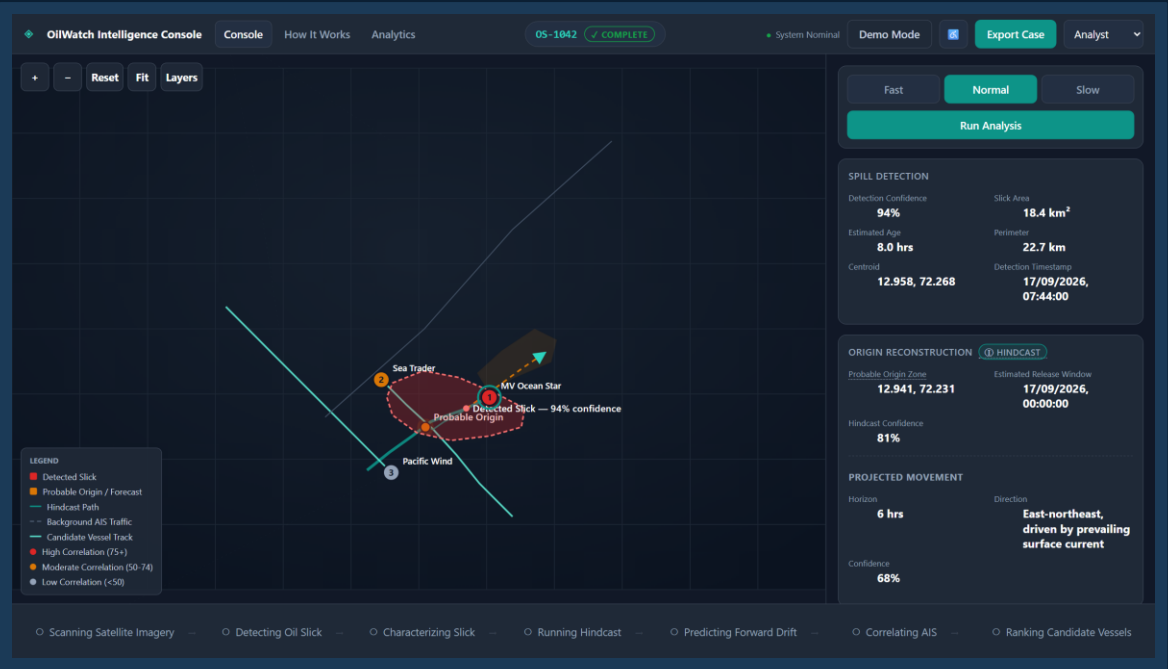


OilWatch

AI-Based Oil Spill & Vessel Detection

From a satellite image to a ranked shortlist of vessels worth investigating, with the evidence shown.

Problem Statement ID	SIH26143
Theme	Miscellaneous
PS Category	Software
Team Name	Mindforgue
Team ID	



The working prototype: OilWatch Intelligence Console

Problem, Idea & Solution

YOUR PROBLEM



Kills marine life

Toxic exposure and heavy oil coating



Damages habitats

Mangroves, coral reefs and coastal ecosystems



Poisons the food chain

From plankton up to fish and larger animals



Hits livelihoods

Fishing waters become unsafe to use

OUR IDEA

One platform that spots an oil spill from satellite imagery, traces it back to its probable origin using ocean-current and wind data, and matches it against AIS vessel tracks to shortlist investigation candidates.

PROPOSED SOLUTION

AI satellite analysis: detects and maps oil slicks automatically

Wind & current modelling: traces the spill to its probable origin and predicts its drift

AIS correlation: matches origin, time and track with nearby vessels

Evidence-based ranking: confidence scores investigators can inspect

Unified GIS dashboard: live view, monitoring and investigation reports

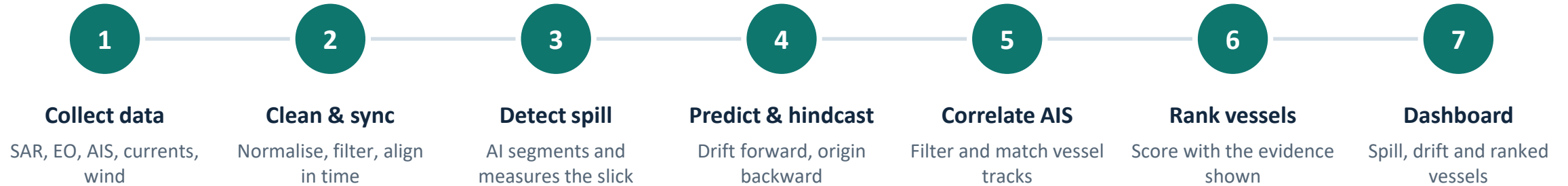
INNOVATION & UNIQUENESS

AI + SAR fusion: spots slicks and cuts false alarms from sea-surface look-alikes

Reverse drift intelligence: wind and currents run backwards to the probable source location and time

Evidence-linked AIS attribution: an explainable score per vessel, built from where, when and how it moved

Technical Approach



SOFTWARE TOOLS

AI & imaging

Python | OpenCV | YOLO / U-Net

Geospatial & storage

QGIS | PostgreSQL + PostGIS | Google Earth Engine

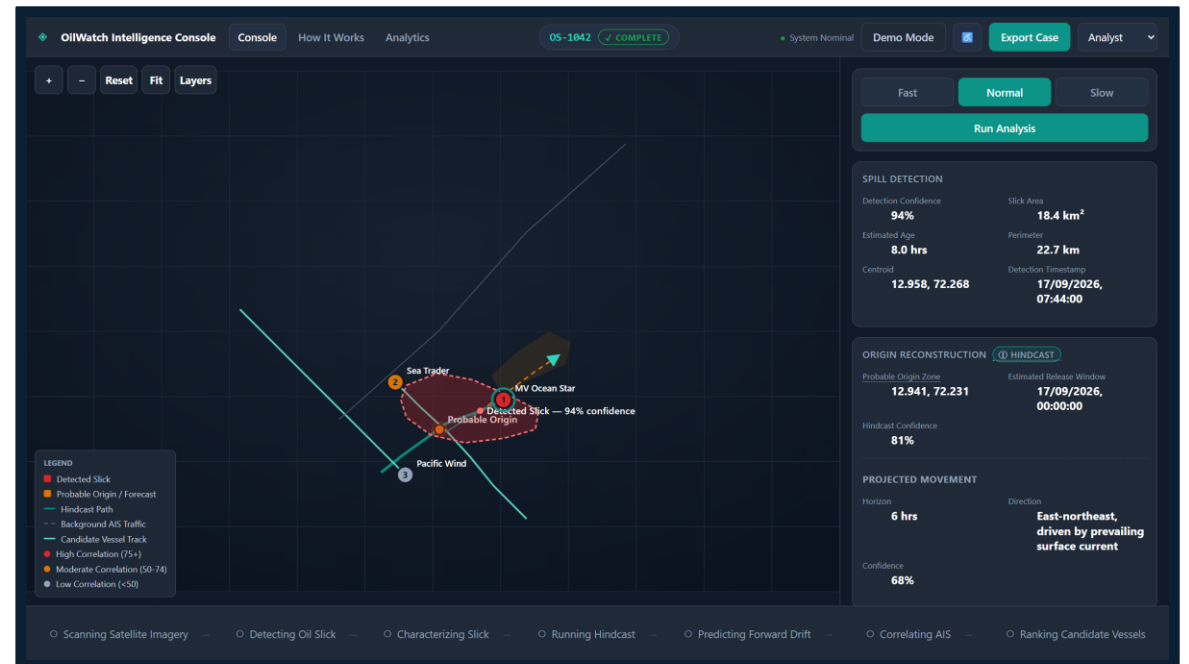
Data sources

Sentinel-1 SAR | AIS datasets | Copernicus Marine / ERA5

App

FastAPI backend | React.js dashboard

Prototype: React + TypeScript, sample data, swappable services



Feasibility and Viability



Technical feasibility

Standard data. Satellite, AIS, wind and current data through open APIs and datasets.

Proven stack. Python, OpenCV, AI/ML and GIS for detection, drift and trajectory matching.

Runs on the web. FastAPI, PostGIS and React. No special hardware.

Already demonstrated. The full workflow runs end to end in our prototype.



Commercial feasibility

Low cost. Built on existing satellite, AIS and open environmental data.

Clear buyers. Coast Guard, ports, maritime and environmental agencies, shipping and insurance firms.

Revenue. Government contracts, annual SaaS subscriptions, data-analysis APIs.

Scales. From one port to national maritime zones.



Challenges & risks

Sea look-alikes. Fuse SAR and EO, and show detection confidence.

AIS gaps or switched-off transponders. AIS continuity is part of every score.

Drift uncertainty. Report a probable origin zone, never an exact point.

Over-reading a score. Framed as investigation support, not proof.

STRATEGY

1 Detect

SAR and EO imagery plus AI find and measure suspected slicks

2 Trace

Wind and currents drift the slick backward and forward

3 Correlate

Rebuild AIS vessel movements around the origin and time

4 Filter & rank

Drop irrelevant vessels, rank by space, time and track

5 Visualize

One dashboard: spill, drift, origin, tracks, ranking

Impact and Benefits

DIRECT IMPACT ON TARGET USERS



Coast Guard & maritime authorities

Faster detection, source tracing and vessel screening



Environmental agencies

Spill extent, drift path and probable origin for quicker response



Ports & shipping authorities

Watch vessel movements with less manual checking



Coastal communities

Quicker action protects fisheries, beaches and coasts



Investigation & regulatory teams

Traceable evidence and standard digital reports

94%

slick detection
confidence

11 to 3

vessels screened down
to ranked candidates

87 / 100

top candidate
attribution score

Prototype demo case OS-1042, synthetic data

STRATEGIC IMPACT

Faster response. Early detection and drift tracking for quick containment.

One intelligence view. Satellite, AIS, ocean and weather data together.

Accountability. Space-and-time evidence behind every candidate.

Scalable. Regional surveillance up to national monitoring.

SOCIAL & ECONOMIC

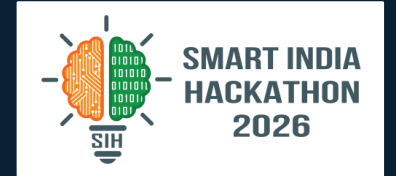
Protects ecosystems and fishing communities.

Safer coasts and less damage to tourism.

Lower cleanup costs through earlier containment.

Less manual work as analysis is automated.

Research and References



EMSA: Oil pollution response

<https://emsa.europa.eu/protecting-the-marine-environment/oil-pollution-response.html>

EMSA: CleanSeaNet detections and feedback data

<https://www.emsa.europa.eu/we-do/surveillance/earthobservationservices/item/4645-cleanseanet-detections-and-feedback-data.html>

IMO: Pollution response

<https://www.imo.org/en/ourwork/environment/pages/pollution-response.aspx>

IMO: Oil spill response

<https://www.imo.org/en/ourwork/environment/pages/oil-spill-response.aspx>

SEE IT RUNNING



Live prototype



Source code

The prototype uses synthetic vessels and sample data. No real vessel is named or implicated, and scores support investigation only.